Differential fluorescence holography

David C. Clark, Myung K. Kim
Digital Holography and Microscopy Laboratory, Department of Physics University of South Florida, 4202 East Fowler Ave, Tampa, FL, USA 33620-5700

ABSTRACT

We present a new technique of three-dimensional tracking by way of incoherent digital holography suitable for non-scanning fluorescence microscopy. By acquiring complex holograms of a sample at different times, a difference hologram can be calculated. This result is another complex hologram representing only the information which has changed throughout the volumetric space during the time interval between one hologram and the next. We first demonstrate the advanced capability of self-interference incoherent digital holography combined with difference holography to track three-dimensional changes in a broadband, unfiltered, sunlit scene containing macroscopic continuous objects. This case is particularly remarkable due to the exceptionally short temporal coherence length and excessive build-up of noninterfering source points. We then demonstrate the ease of adaptation to microscopy, successfully converting a standard two-dimensional commercial microscope into a powerful three-dimensional tool. By overcoming these challenges, we show the feasibility and ease at which this technique may be adapted to the versatile, functional imaging of fluorescence microscopy.

Keywords: SIDH, differential imaging, fluorescence microscopy, incoherent holography, three-dimensional tracking

1. INTRODUCTION

Fluorescence microscopy is an indispensable imaging tool in modern biomedical research. Without the use of scanning techniques, however, only two-dimensional (2D) images are captured. In contrast, holography records sufficient information to recreate the three-dimensional (3D) optical field emanating from the scene, including both amplitude and phase[1]. The three-dimensional recording is made possible by the interference of the object's optical field and a reference optical field, and therefore requires coherence between the two. In D. Gabor's original conception, the reference is realized from a part of the illumination undisturbed by the object[2]. The invention of the laser made it possible to provide a coherent reference field explicitly[3]. Because fluorescence imaging is inherently incoherent, holography has been largely precluded from fluorescence until recent advancements.

The introduction of Fresnel incoherent correlation holography (FINCH) has fully demonstrated the ability to generate holographic images of incoherent object fields, including 3D fluorescence microscopy[4,5]. In this technique, two copies of the object wave are superposed with different phase factors imposed by a spatial light modulator (SLM), so that every point source on the object produces the Fresnel zone interference pattern[6]. We have since developed a technique conceptually similar to FINCH called self-interference incoherent digital holography (SIDH), in which the SLM is replaced by a modified Michaelson interferometer with a differential phase curvature applied by its two mirrors[7]. For continuous objects or scenes, the spatial incoherence of the object points leads to rapid build-up of incoherent background which is removed by standard phase-shifting digital holography[8]. The resultant hologram can then be numerically propagated to any distance to reconstruct the desired optical field. Figure 1 shows an example of a broadband sunlit holographic scene successfully recorded by our SIDH camera using lensing and numerical aperture such that planes of interest are visibly distinguishable. Because coherence length is inversely proportional to bandwidth, this presents a notable challenge which is overcome by our SIDH process.

We introduce here a new process of differential self-interference incoherent digital holography (diff-SIDH) and demonstrate its highly effective capabilities and ease of adaptation to fluorescence microscopy. We first fully demonstrate the process on photo-quality holograms of unfiltered broadband natural sunlit objects. We then demonstrate the easy adaptation to commercial grade microscopy equipment, successfully converting a professional 2D imaging instrument into a powerful 3D imaging and tracking tool.

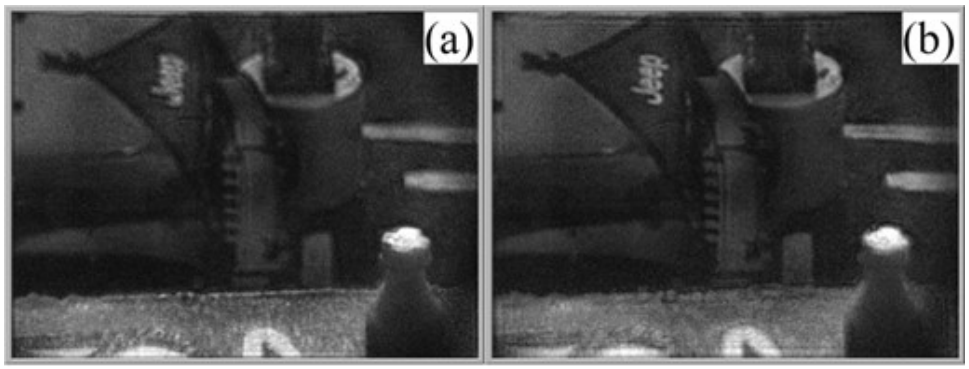

Figure 1. A photo-holographic image of a 1 cm tall rook in a window sill 30 cm away and a vehicle (Jeep) in an outside, lower-level, parking lot 3700 cm away. The hologram is first propagated to the rook's focus plane (a), then to the Jeep's focus plane (b).

2. EXPERIMENTS

A basic SIDH apparatus is represented in figure 2. It consists of the input lens, the relay lens, the interferometer, and a CCD array. The input lens produces an intermediate image as the input for the SIDH module. The relay lens is selected and positioned to create an appropriate final image space relative to the CCD plane (focal planes of interest should not fall directly on the recording plane for proper self-interference). The interferometer consists of a beam-splitting cube and two mirrors. One is a plane mirror mounted on a piezo-actuator, which is driven by a 5 volt ramp signal for phase-shifting. The other is a curved mirror to generate differential phase curvature.

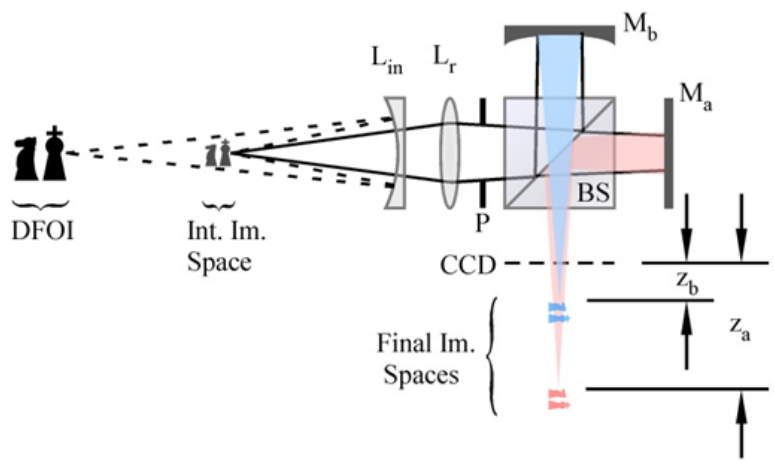

Figure 2. Experimental diagram for diff-SIDH. An input lens, L_{in}, projects objects in the depth field of view, DFOI, into the intermediate image space as input to the SIDH module. The SIDH module consists of a relay lens, L_r, an iris, P, a beam splitter, BS, a flat piezo-mounted mirror, M_a, a curved mirror, M_b, and a CCD camera.

To ensure interference, the distances of the two mirrors are matched to within a margin based on the temporal coherence of the desired light source[9]. The degree of this adjustment is visibly verified by imaging a single incoherent, out-of-focus, LED source for best fringe contrast. Then, several exposures of an incoherent illuminated scene are captured while the phase-shifting piezo-mounted mirror travels through at least one full 2π phase excursion. The complex hologram is calculated from these N intensity profiles, I_n, by

$$H = \frac{1}{N}\sum_{n=0}^{N-1} I_n \, \mathrm{e}^{i2\pi\frac{n}{N}} \ . \tag{1}$$

It is this zero-plane (CCD-plane) complex hologram, now devoid of DC and virtual twin images, that is used for further processing, such as diff-SIDH or propagation to reconstruction planes. A minimum of $N = 4$ is required to eliminate the unwanted terms; however, we chose $N = 20$ for all holographic images presented here to ensure removal of DC signal and smooth random noise effects. While a detailed description of the reconstruction is available in [7], we will summarize here that our resulting complex holograms are simply propagated, as per diffraction theory, by desired distances, z_h, described by

$$z_h = -\frac{z_a z_b}{z_a - z_b} \tag{2}$$

where $z_{a,b}$ are the axial positions of the final focal planes of a desired object plane through the two paths of the interferometer and are measured relative to the CCD plane. For a well-defined system, these values are easily translated from object space to final image space and back again. For our purposes here, we use thin lens approximated geometry, as well as compensation for image shift due to the 25.4 mm BK-7 beam cube.

2.1 Broadband differential incoherent holography

The process of diff-SIDH is applied here to the case of chess pieces on a chess board illuminated by sunlight through an office window. This experiment is summarized in figure 3. The knight is moved from its initial position in figure 3(a) to a final position shown in figure 3(b and c). The difference hologram is calculated from the un-propagated initial and final complex holograms,

$$\Delta H = H_i - H_f \ . \tag{3}$$

The difference hologram for this experiment is shown in figure 3(d-f) and contains the 3D optical field information of the knight in both initial and final positions, while the unchanged information, including the stationary king, are absent. As is shown, we shift the focus of each hologram from back knight to king to forward knight positions (figure 3(a-c), respectively) and this information is preserved in the difference hologram (figure 3(d-f), respectively). It can also be noticed that the information on the chess board, both overlap and specular reflection of the knight, is also preserved in a difference hologram, since this represents changed information as well.

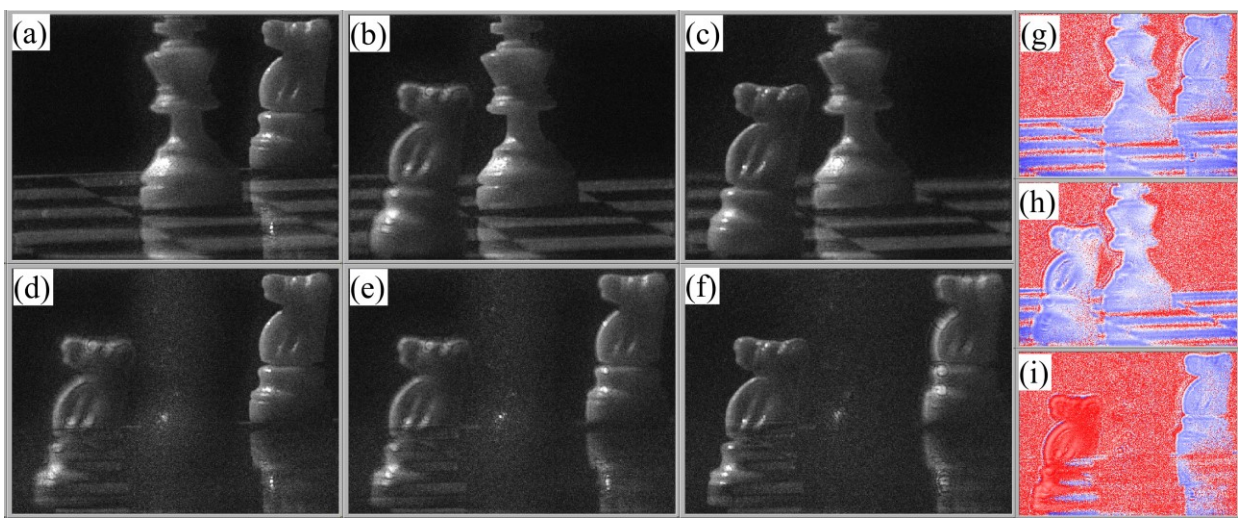

Figure 3. Summary of broadband differential incoherent holography experiment. Chess pieces, knight and king, were arranged as shown in the initial hologram, (a). The knight was then moved to a new position and a final hologram was recorded, (b) and (c). A difference hologram was computed from these and appears in (d), (e), and (f). Holograms (a) and (d) are propagated to the back knight plane, (b) and (e) are propagated to the king plane, and (c) and (f) are propagated to the front knight plane. (g), (h), and (i) are the phase images of the initial, final, and differential holograms showing a relative shift in the profile representing the final knight position.

The amplitude images displayed in figure 3(e-f), by nature, mask the information which describes the order of starting and ending positions in the differential image. Of course, this information is retained in this complex field and can be visualized by displaying the separate complex components or the phase maps of the associated holograms as shown in figure 3(g-i) where the new information imposed by the subtrahend, such as the new knight position, is shifted in the difference hologram, while the information on the original knight position is un-shifted.

From the difference hologram, the overall change of the scene is computed and movement of objects in the scene is tracked in three dimensions. In the present case, if we select the eye of the knight as an easy point of reference, a 3D displacement vector can be calculated showing the projected path between the time of the initial and final holograms. Using the plane parallel to the CCD plane that contains the initial eye as our reference and setting the origin at the center of our field of view (FOV), position vectors can be constructed from our difference hologram for the initial and final eye positions by

$$\mathbf{v}_i = [\frac{px_i}{nx} x_i, \frac{py_i}{ny} y_i, z_i]$$

where px and py are the lateral pixel position coordinates, nx and ny are the lateral FOV's in pixels, and x_i and y_i represent the object space FOV's in units of length. This object space FOV is, of course, scaled based on the depth plane of interest at z_i by the relation

$$x_i = x_0 \left(\frac{z_0 - z_i}{z_0} \right)$$

(and identically for y_i) where x_0 is the known object space FOV at a plane located at z_0, and z_i is measured from this z_0 plane and is translated from z_h as described in the following paragraph. For our example, we have

$$\begin{aligned} \Delta\mathbf{v} &= \mathbf{v}_2 - \mathbf{v}_1 \\ &= [-5.4, 2.5, 62] \text{ mm} - [8.9, 7.1, 0] \text{ mm} \\ &= [-14.3, -4.6, 62] \text{ mm} \end{aligned}$$

for the knight's displacement vector in object space. Resolution of the z-component of this vector is related to numerical aperture and will be discussed shortly.

It is important to consider the relative locations of subjects of interest, i.e., the depth field of interest (DFOI). Since we are interested in tracking objects in a 3D volume of space, our intermediate image space should consist of an appropriate volume of space as input for the diff-SIDH system. In the above experiment, the DFOI (spanning the length of the chess board) is between 350 mm and 430 mm from the relay lens. By thin lens geometry, this translates to a 2.2 mm range in intermediate image space using a negative 75 mm input lens placed 7 mm in front of the relay lens. Following similarly, the positive 50 mm relay lens of the SIDH module translates this input to a range of z_a (plane mirror path) and z_b (curved mirror path) values of 25.7 to 39.8 mm and 7.62 to 17.4 mm, respectively. Thus applying Eq.(2), we expect objects in our DFOI to come into focus somewhere within the propagation range of -10.8 to -30.8 mm. Of course, these relationships are non-linear and a numerical simulation of our constructed system produces a translation from object space DFOI into final holographic propagation space as plotted in figure 4(a). Specifically, our three object positions of interest consisting of the 425 mm back knight position, the 401 mm king position, and the 363 mm forward knight position are predicted to be in focus at propagation distances; -11.7, -16.3, and -26.4 mm, respectively. These three predicted propagation planes were chosen for display in figure 3(a and d), 3(b and e), and 3(c and f), respectively. Clearly, these are appropriate planes for the objects in question.

While choice of lensing is quite important for defining the DFOI for a given application, it is the numerical aperture (NA) that will ultimately determine the systems depth resolution. We are able to see, qualitatively, that our holograms can distinguish between the object planes of interest due to a degree of blur that exists in the out-of-focus planes. This blur is described by a parameter known as the circle of confusion, c, which is proportional to NA by

$$c = \frac{2NA|z - z_0| f}{z}, \tag{3}$$

where f is the lens focal length, z_0 is the position of the in-focus plane, and z is the position of an out-of-focus plane. The expected blur in units of pixels vs. object space for a standard (non-holographic) image of our DFOI with the king plane set to in-focus is plotted in figure 4(b). A qualitative comparison with figure 3(b and e), which are numerically propagated to this plane, shows that the relationship here is reasonable, though likely a holographic analog of the blur relationship would be necessary for a more detailed quantitative analysis. For now, we will use this analogy with standard imaging to describe the general importance of such a parameter in defining depth resolution. If we consider a generic focus algorithm for digital imaging, it will be limited in some way by units of pixels such as we have used here. For the sake of our qualitative comparison, we may assign the minimum of one pixel of blur before we detect an edge to be out of focus, thus our depth resolution would be 4.9 mm in object space for the region about the king plane. Note that because the NA is different for different object points, the depth resolution will also change depending on the target within the hologram.

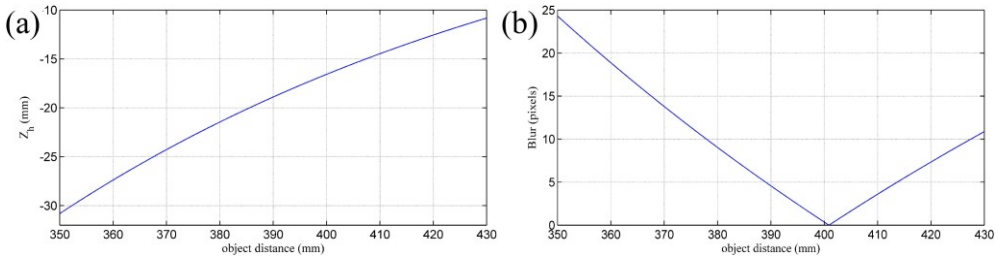

Figure 4. Space and blur predictions. (a) Numerical simulation showing the holographic propagation distances, z_h, expected for object planes within the DFOI. (b) Expected image blur in pixels for objects in the DFOI when the king plane (401 mm away) is brought into focus.

2.2 Differential incoherent holographic microscopy

In preparation for working with fluorescence microscopy materials and instruments, we have fabricated a similar system to that above (figure 2) which we mount directly to the camera port of a standard commercial microscope. In essence, the input lens is replaced by the microscope as the input device to create the intermediate image space. In this way, the bright-field microscope images are successfully recorded here as complex holograms containing the 3D volumetric information of the object space extending above and below the microscope's normal input plane.

A slide containing a USAF1951 resolution target was placed on the microscope stage while a coverslip containing two particle-like structures was held approximately 8 mm above by a separate translation stage. The volume space was illuminated by a single color red LED (notably similar in incoherence and bandwidth to typical fluorescence emittance) and an initial hologram was recorded. The two-particle system was then translated laterally and a final hologram was recorded. Just as above, a difference hologram was calculated resulting in a hologram containing only the two-particle system in its initial and final positions while all unchanged information from all planes is absent.

A summary of this experiment appears in figure 5. Figure 5(a-c) shows the amplitude representation of the initial, final, and difference holograms, respectively, all propagated to the resolution target plane. It is obvious in the difference hologram that this plane is now devoid of the resolution target information and that only the out-of-focus projections of the translated two-particle system are present. Meanwhile, figure 5(d-f) contain these same three amplitude holograms, but now propagated to the particle plane. At this plane, the two-particle system is clearly identified, in-focus, in the difference hologram at both its initial and final positions. As discussed in the previous experiment, a shift of complex information describes the initial and final positions of moved objects, although this is hidden in an amplitude representation. In this case, we display in figure 5(g-i) only the imaginary component of each of these same holograms and propagate them once again to the particle plane. In this representation, it is easy to detect the two particles, identified by circles for convenience, in their initial positions in figure 5(g), and their final positions in figure 5(h). More importantly, in figure 5(i), the final position of this two-particle system is shifted in this representation, clearly distinguishing it from the un-shifted initial position.

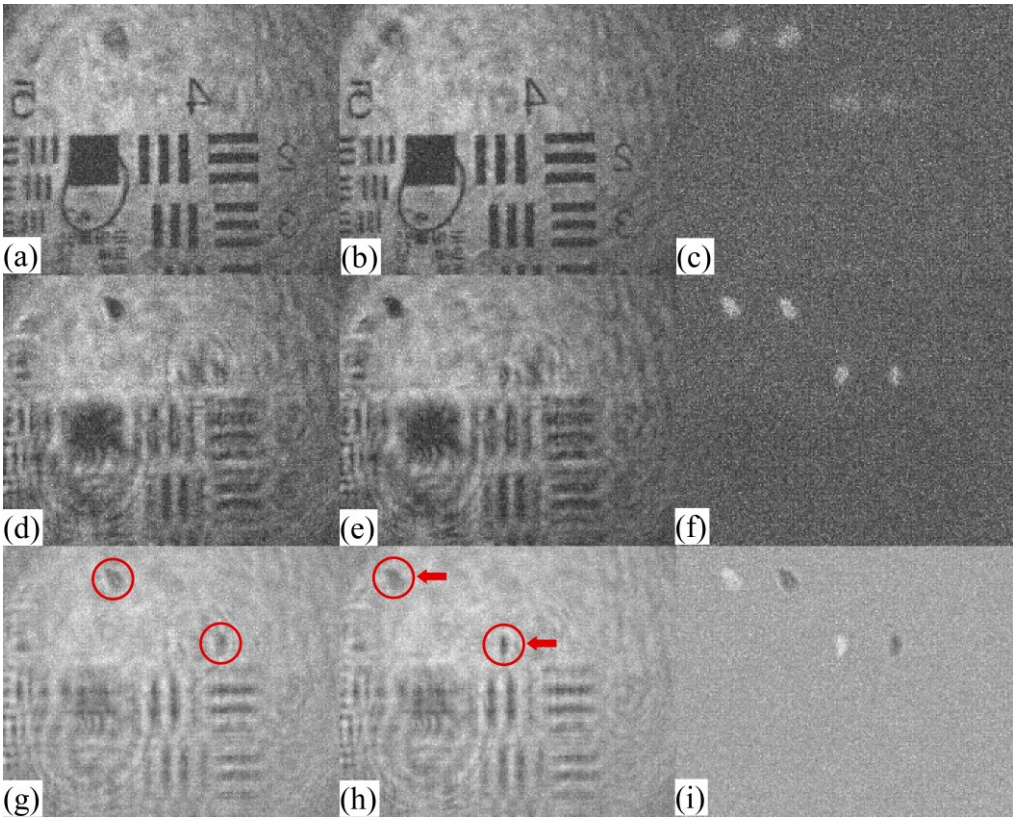

Figure 5. Summary of differential incoherent holographic microscopy experiment. (a), (b), and (c) are the initial, final, and differential holograms, respectively, all propagated to the resolution target plane. (d), (e), and (f) are the initial, final, and differential holograms propagated to the particle plane. (g), (h), and (i) are the imaginary component images of the initial, final, and differential holograms at the particle plane showing a relative shift (to white) in the profile representing the final particle positions.

3. CONCLUSION

We have demonstrated the full capabilities of diff-SIDH for the most challenging case of broadband, sunlit, continuous objects using photo-quality 3D holograms captured by our SIDH camera. Successfully overcoming the remarkably short temporal coherence and the excessive incoherent background build-up of such cases suggests that the process will be highly effective in the far less challenging case of fluorescence applications. Fluorescence microscopy, for example, involves narrow-band emission and often point-source-like structures, which will be much less constraining on the SIDH module promoting substantially higher rates of holographic capture. We have furthermore demonstrated that our modular design allows for easy implementation into existing professional grade microscopes, converting existing 2D imaging instruments into powerful 3D holographic tools. We now propose to develop a holographic fluorescence microscope that combines the three-dimensional imaging and tracking capabilities of diff-SIDH with the versatile functional imaging by fluorescence microscopy.

In addition, we are currently developing a single-shot, off-axis SIDH method which will eliminate the need for phase-shifting by incorporating a slight tilt between the self-interfering copies[10]. This may be quite significant for dynamic biological applications where temporal resolution is important. We expect this to be especially effective for fluorescence applications.

REFERENCES

[1] Hariharan, P., [Optical Holography: Principles, Techniques, and Applications], Cambridge University, (1996).

[2] Gabor, D., "A new microscopic principle," Nature 161(4098), 777–778 (1948).

[3] Leith, E. N. and Upatnieks, J., "Wavefront reconstruction with continuous-tone objects," J. Opt. Soc. Am. 53, 1377–1381 (1963).

[4] Rosen, J. and Brooker, G., "Non-scanning motionless fluorescence three-dimensional holographic microscopy," Nat. Photonics 2, 190–195 (2008).

[5] Rosen, J. and Brooker, G., "Fluorescence incoherent color holography," Opt. Express 15, 2244–2250 (2007).

[6] Rosen, J. and Brooker, G., "Digital spatially incoherent Fresnel holography," Opt. Lett. 32, 912–914 (2007).

[7] Kim, M. K., "Incoherent digital holographic adaptive optics," Appl. Opt. 52, A117–A130 (2013).

[8] Yamaguchi, I. and Zhang, T., "Phase-shifting digital holography," Opt. Lett. 22, 1268–1270 (1997).

[9] Dubois, F., Callens, N., Yourassowsky, C., Hoyos, M., Kurowski, P., and Monnom, O., "Digital holographic microscopy with reduced spatial coherence for three-dimensional particle flow analysis," Appl. Opt. 45, 864–871 (2006).

[10] Hong, J. and Kim, M. K.. "Single-shot self-interference incoherent digital holography using off-axis configuration," Opt. Lett. 38, 5196-5199 (2013).